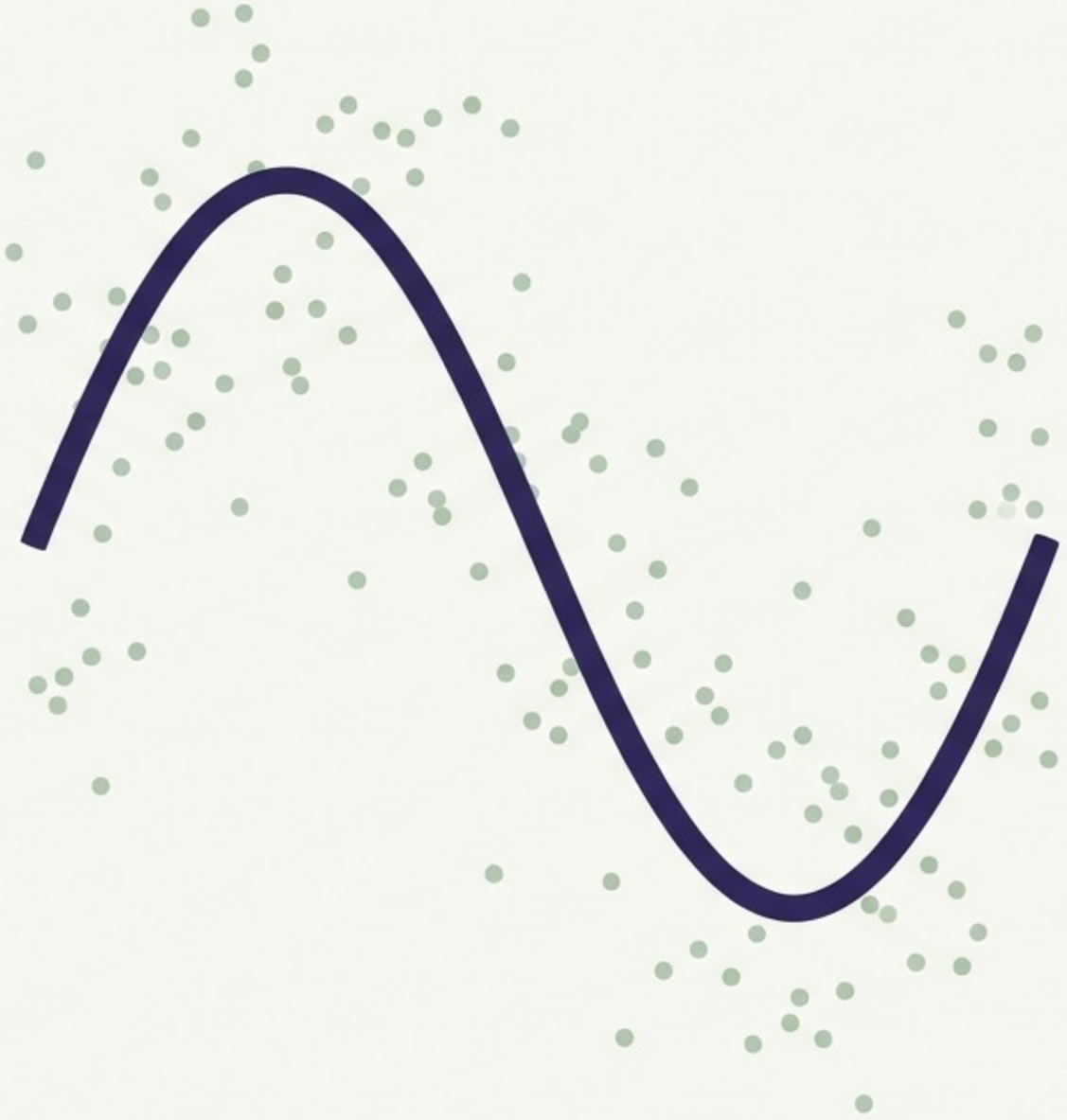
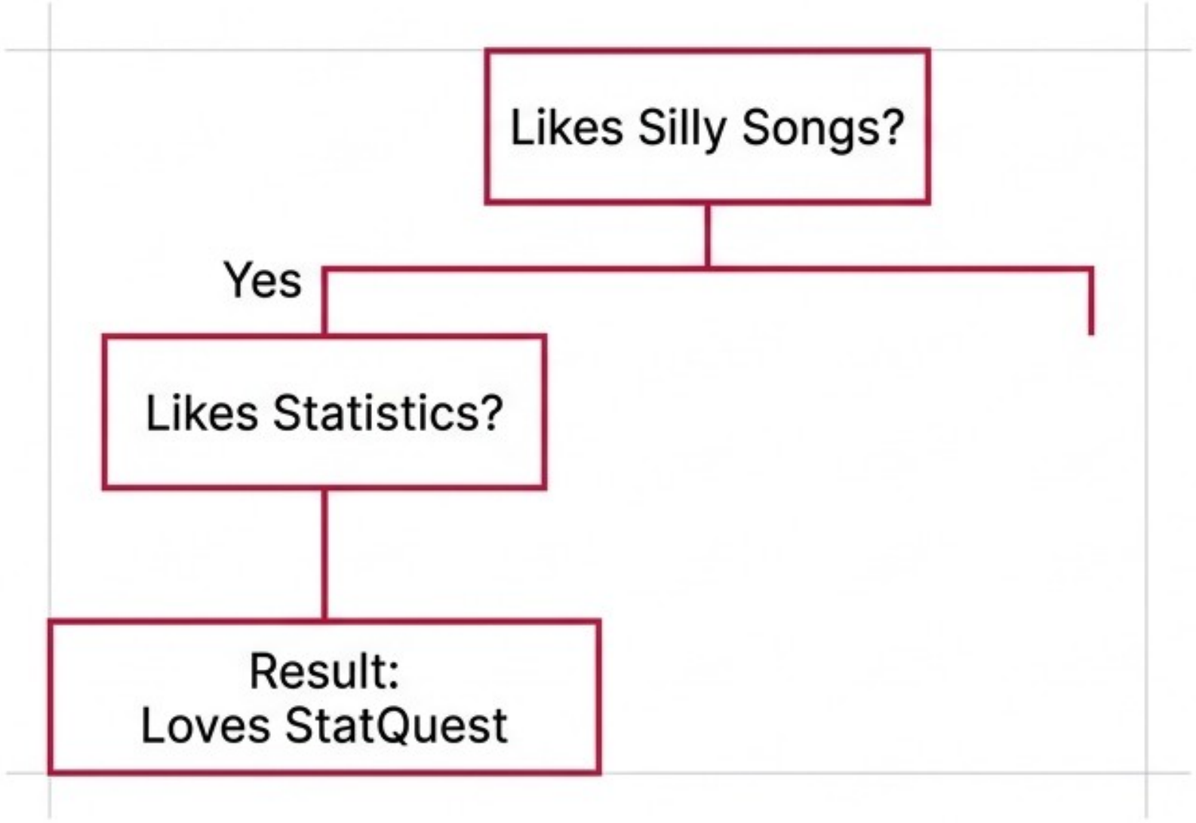
Module 5: Machine Learning Fundamentals

Finding the Signal in the Noise



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Categorising and Predicting the Future





Classifications

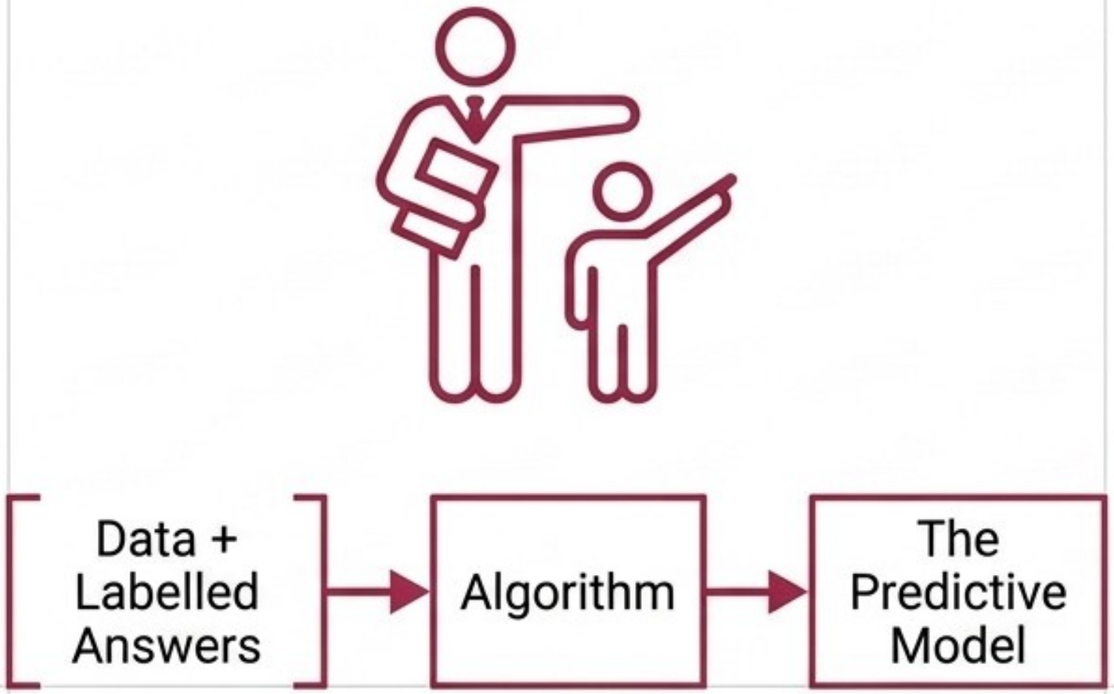
Sorting data into distinct groups based on historical patterns (e.g., Will they like this?).

Predictions

Forecasting continuous numerical outcomes based on established trends (e.g., How fast will they run?).

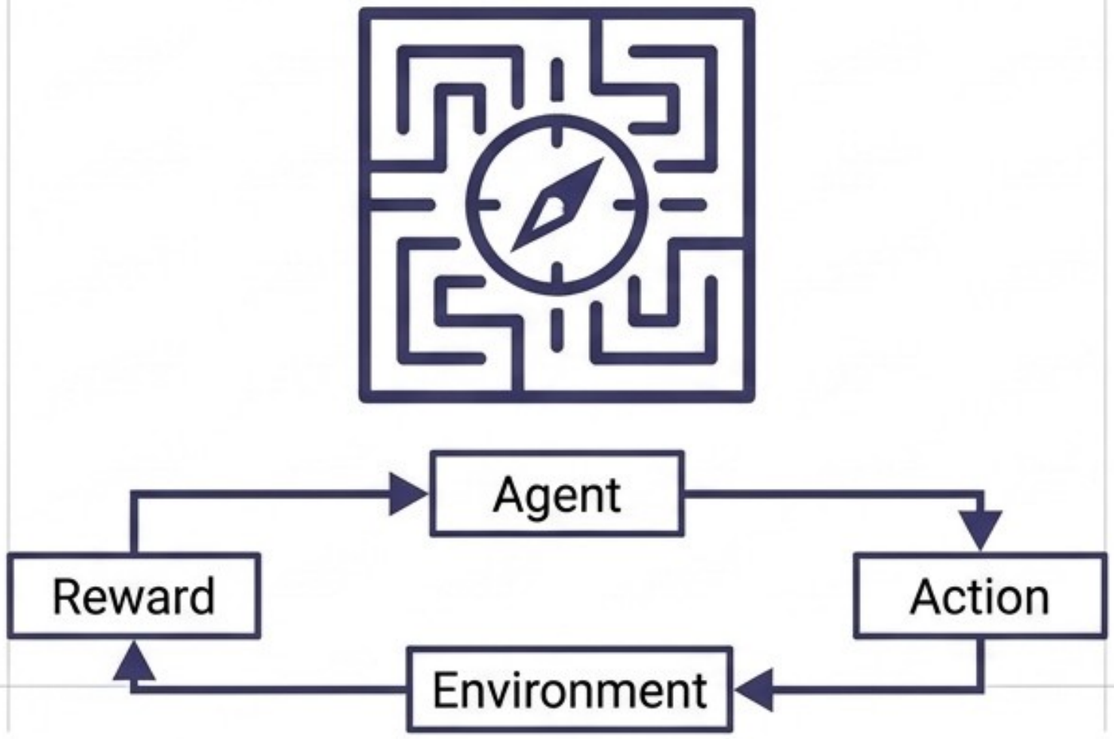
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Two Distinct Paths to Artificial Intelligence



Supervised Learning

Learning from historical, labelled data to map inputs to specific, known outputs.



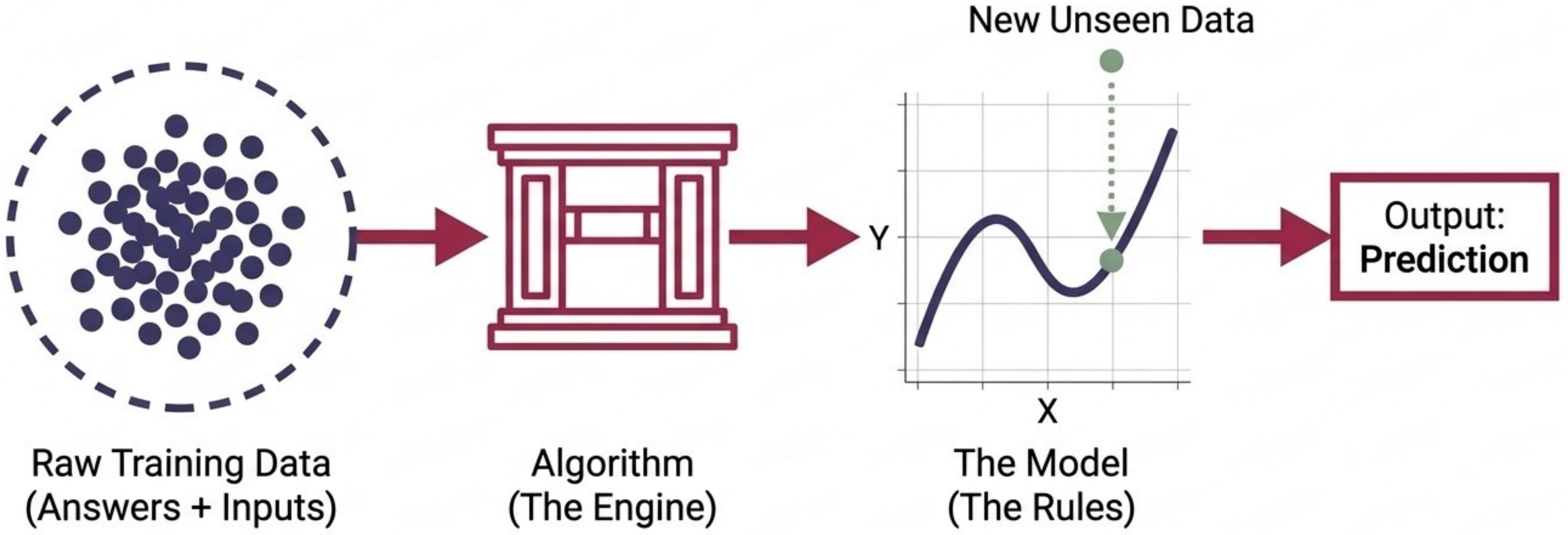
Reinforcement Learning

Learning dynamically through trial, error, and adaptation within a responsive environment.

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The Supervised Learning Pipeline

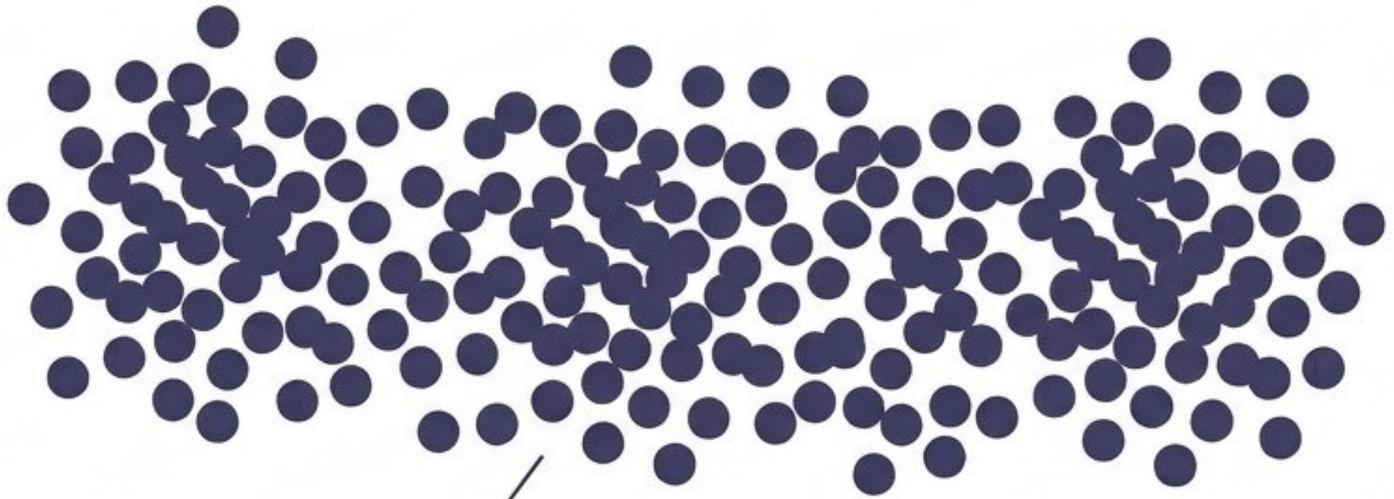
In traditional programming, we input rules to get answers. In Machine Learning, we input answers and data to let the algorithm generate the rules.



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The Crucial Split: Training vs. Testing

Training Data (75%)



Used exclusively to estimate parameters and build the model.

Testing Data (25%)

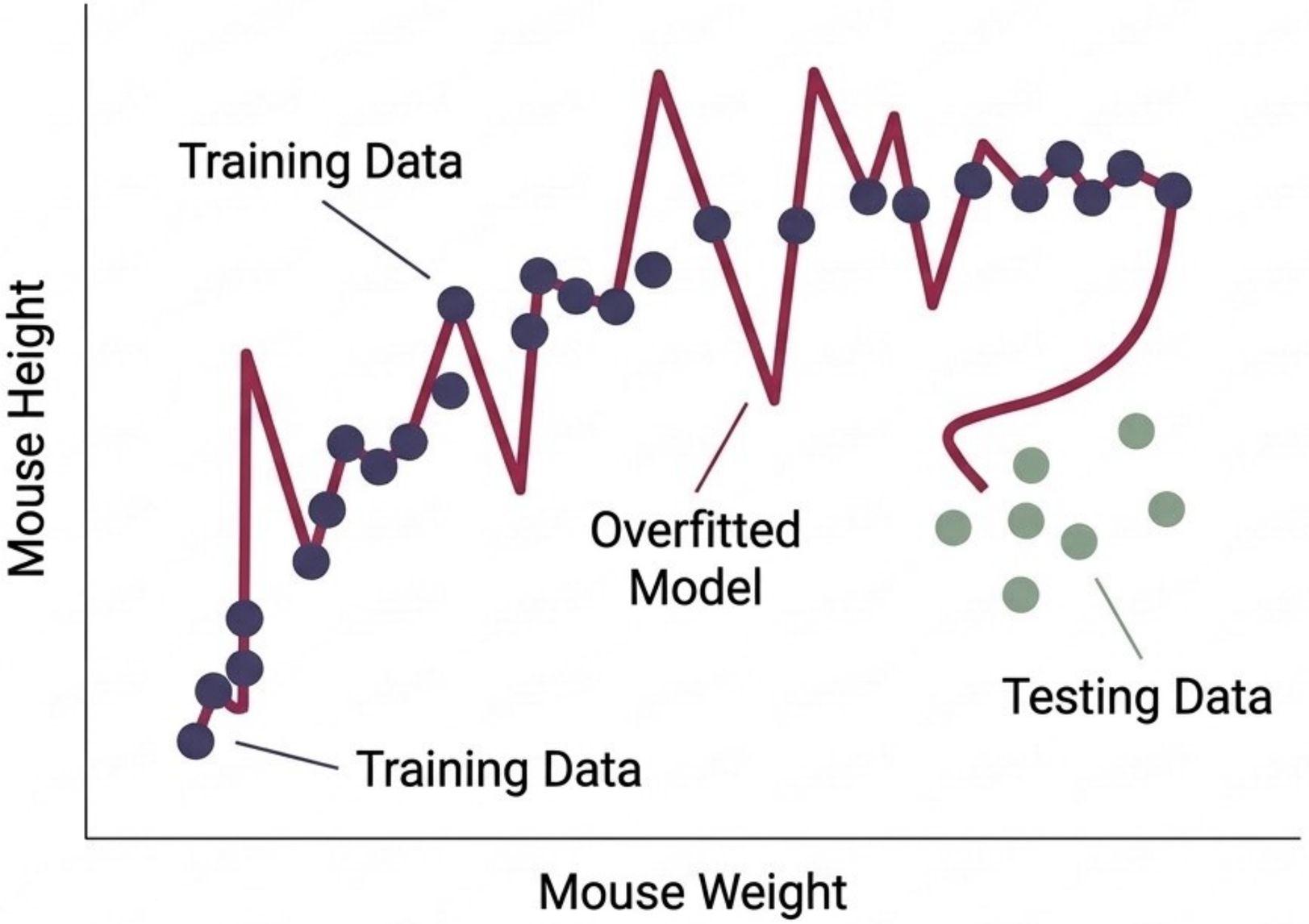


Used only to evaluate how well the model predicts reality.

The Cardinal Sin: Never use your testing data to train your algorithm. We must test the model on data it has never seen before.

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The Illusion of Perfection (Overfitting)



The Trap:

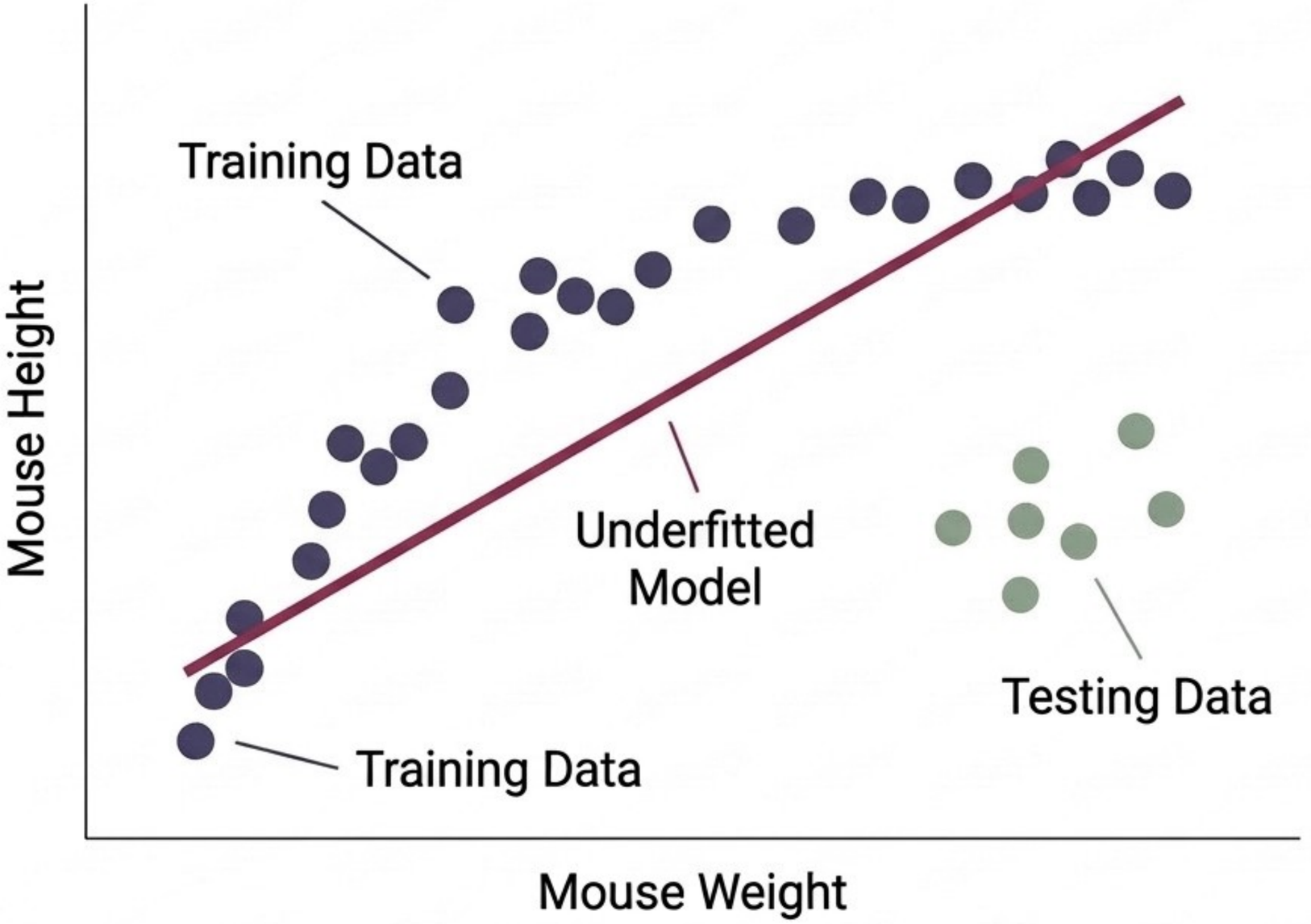
The model memorises the training data's noise rather than learning the actual underlying relationship.

Symptom:

High Variance. It looks perfect in training, but its performance is wildly unpredictable when exposed to new data.

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The Danger of Oversimplification (Underfitting)



The Trap:

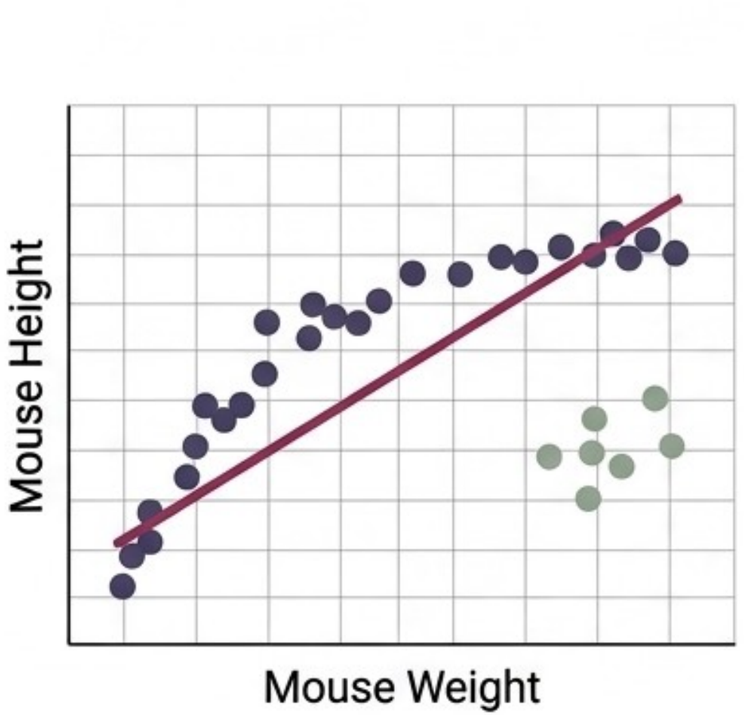
The model is too simple and lacks the flexibility to capture the true mathematical relationship.

Symptom:

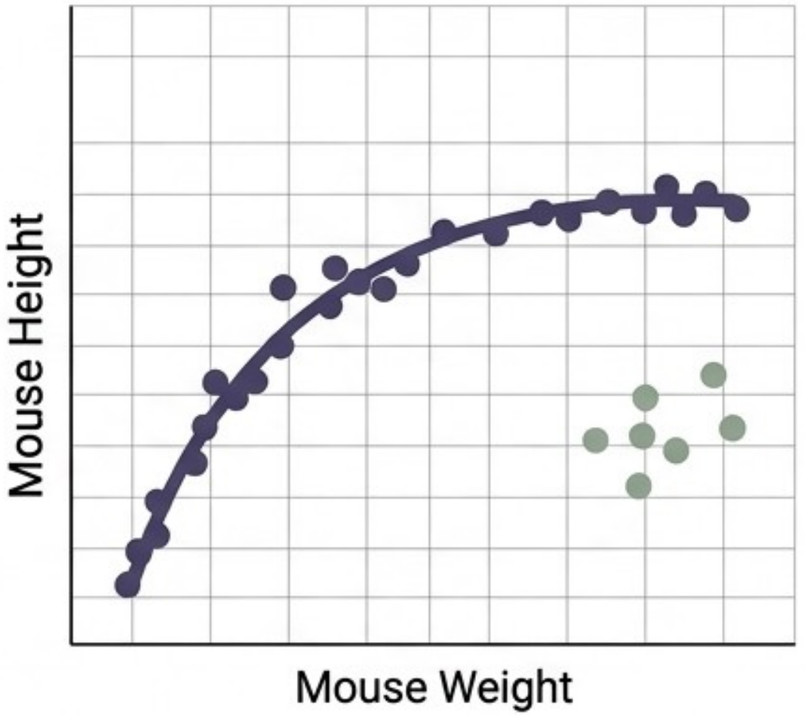
High Bias. It is consistently inaccurate, regardless of which data set it is tested against.

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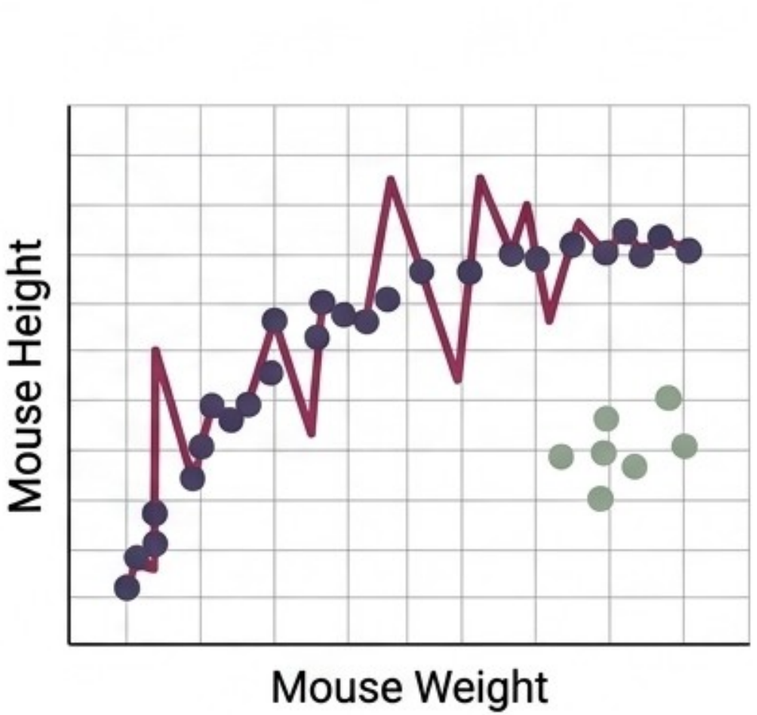
The Bias-Variance Tradeoff



High Bias (Underfit)
Fails to capture the trend.



The Sweet Spot
Balanced complexity.

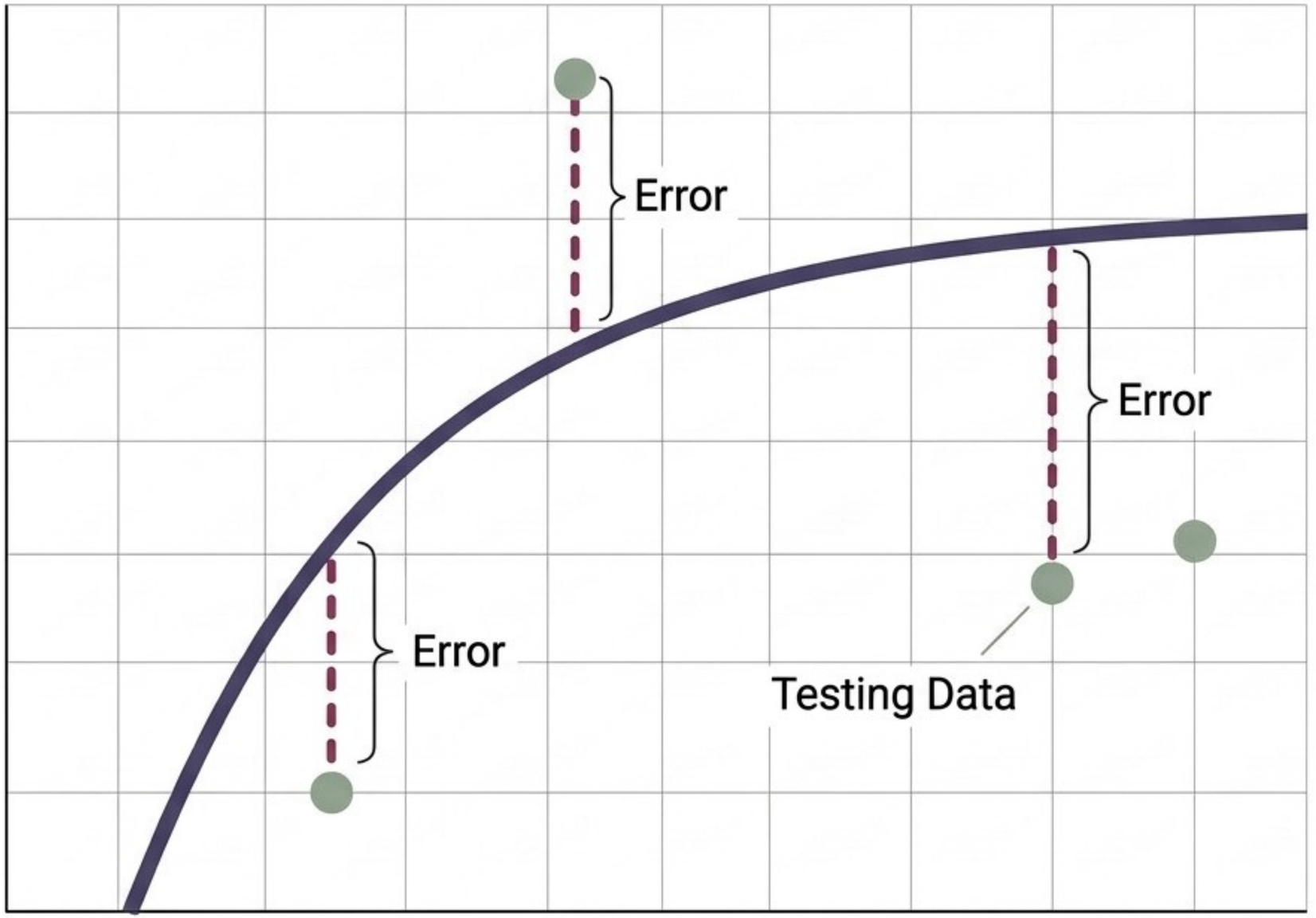


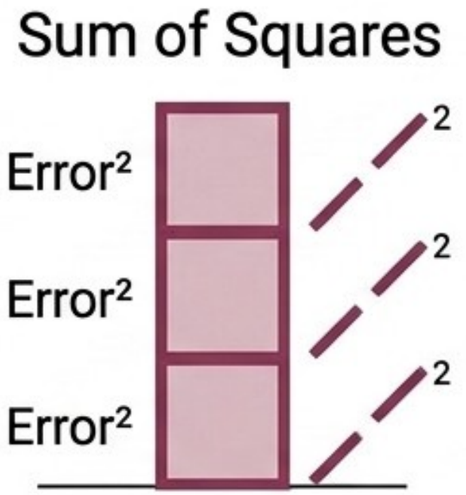
High Variance (Overfit)
Memorises the noise.

The goal of Machine Learning is finding the sweet spot: low bias (accurate to the true relationship) and low variance (consistent and reliable across new data sets).

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Measuring the Distance from Reality



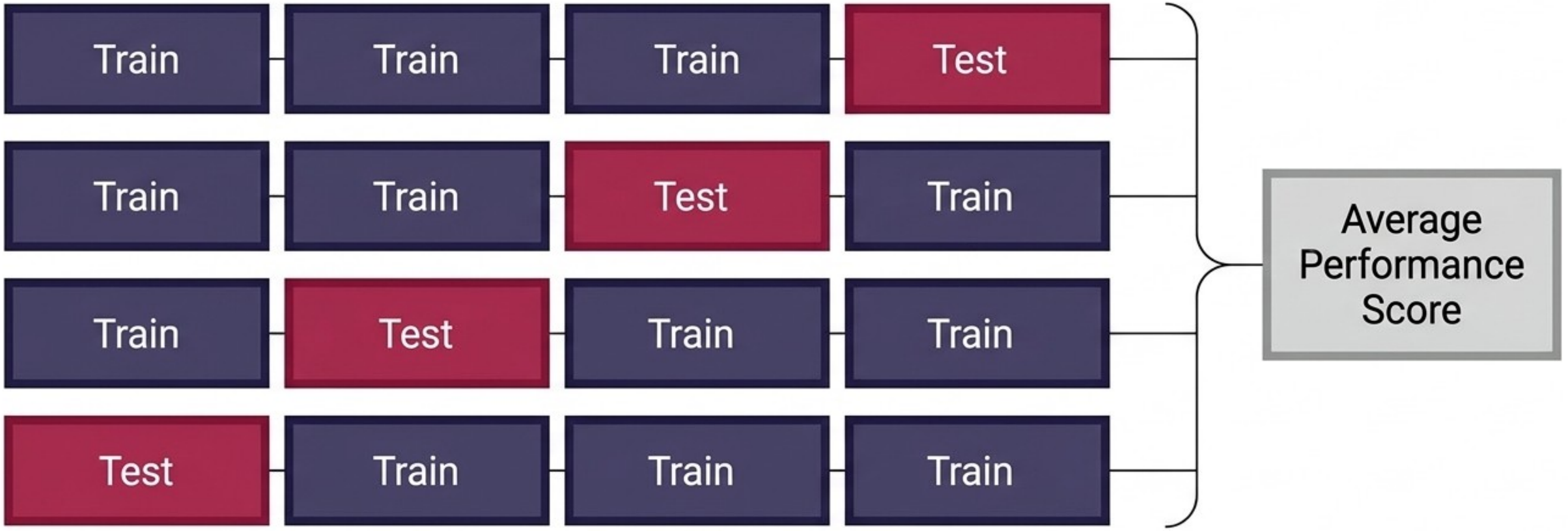


The Method: We evaluate models by measuring the exact physical distance between the predicted reality (the curve) and the actual reality (the testing data).

The Rule: The model with the smallest total error on the testing data wins.

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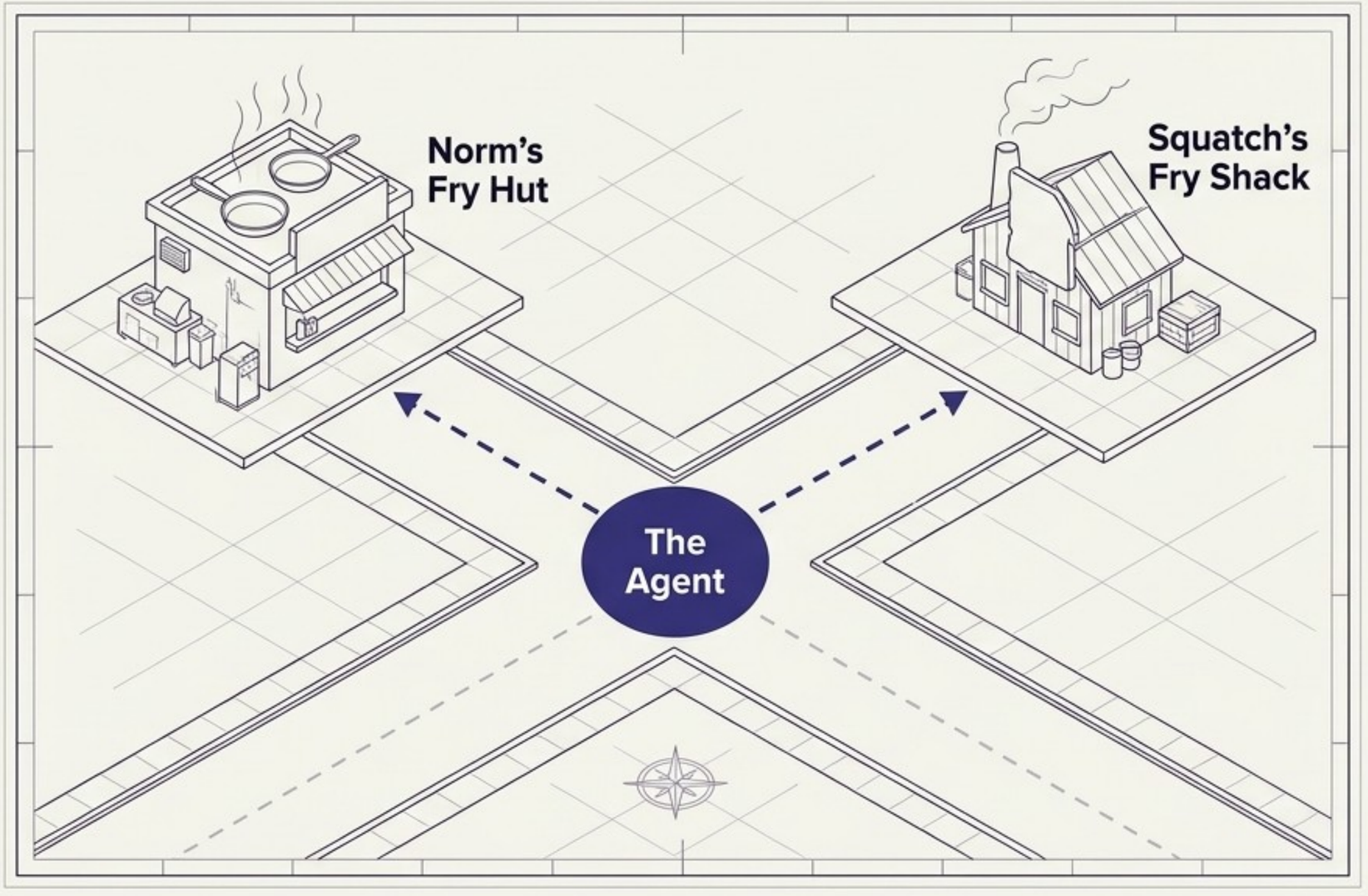
Maximising Data with K-Fold Cross-Validation



- Don’t just test once. Divide data into K blocks (e.g., 4-fold or 10-fold).
- Rotate which block acts as the testing data until every single sample has been tested.
- This extracts maximum confidence without violating the rule of mixing train and test data.

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A Paradigm Shift: Reinforcement Learning



No labelled data. No predetermined lines. No historical datasets.

The model learns dynamically from scratch by interacting with an environment and experiencing consequences firsthand.

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The Mechanics of Experience



Agent
The decision-maker
(The Algorithm).

Action / Policy
The mathematical probability
guiding the choice.

Environment
The world being
navigated.

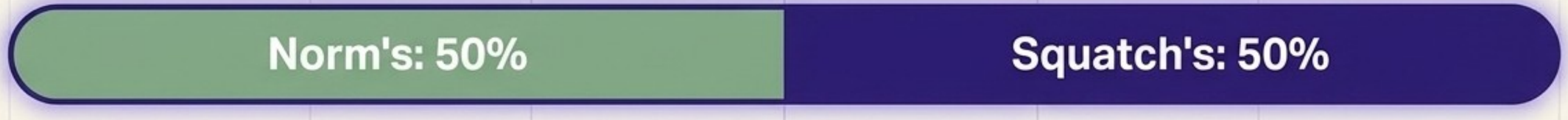
Reward
The feedback score
(1 for success, 0 for failure).

The Agent explores the Environment, takes an Action dictated by its Policy, receives a Reward, and uses that Reward to update the Policy for the next loop.

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Adapting the Policy in Real Time

State 1 (Baseline)



The Experience



Reward = 1
(Satisfying Order at Norm's)



Applying Learning Rate (0.1)

State 2 (Updated Policy)

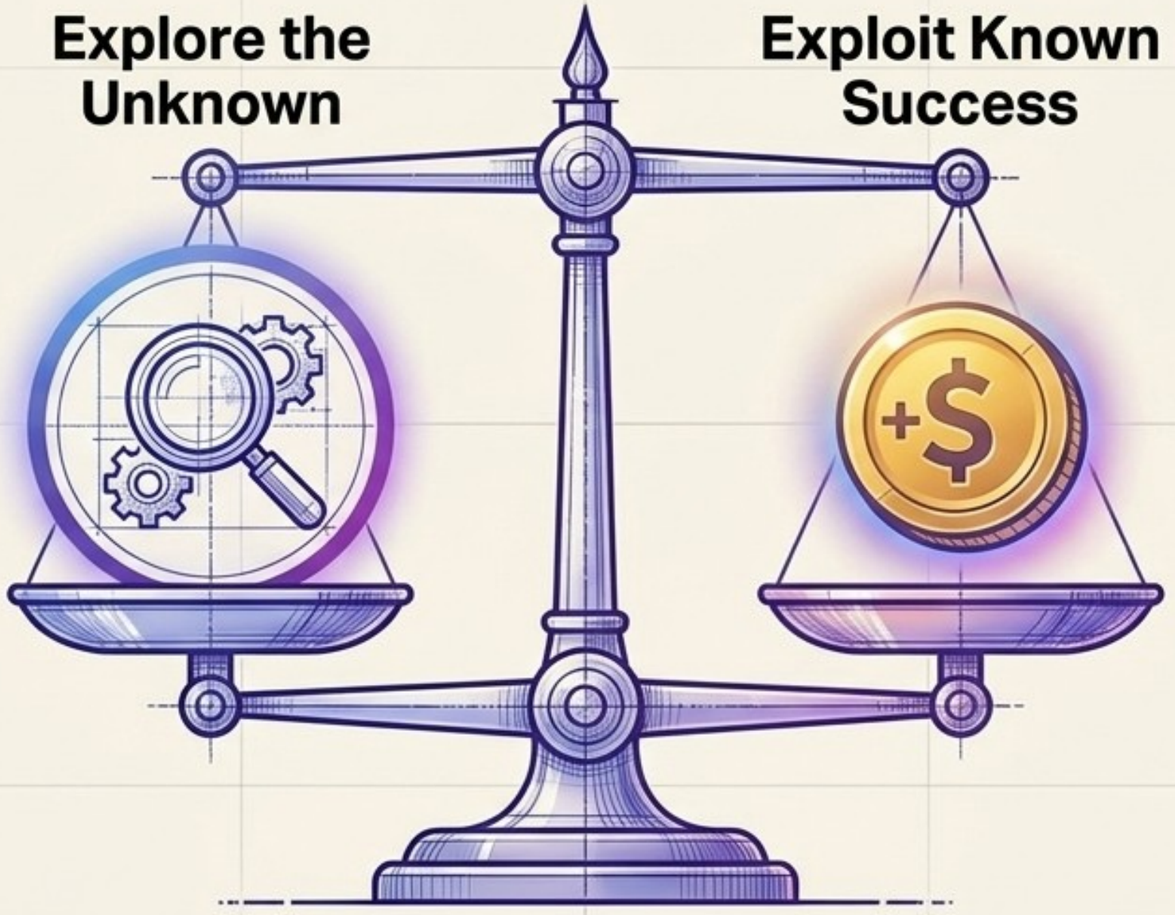


The Update: Based on the Reward, the model updates its Policy to increase the probability of making the successful choice again.

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The Dilemma: Exploration vs. Exploitation

Why not update the probability to 100% immediately?



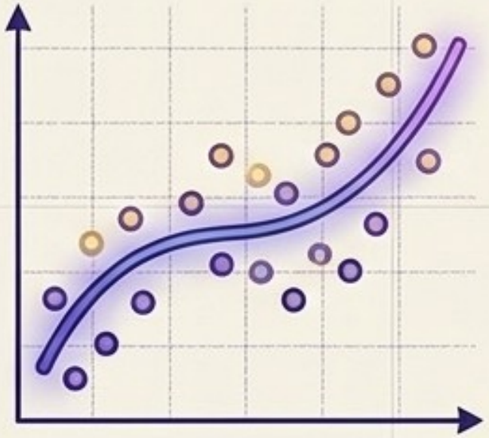
If we jump to 100% immediately, we cease learning.

The algorithm must constantly balance exploiting known good outcomes with exploring to ensure conditions haven't changed or to find a hidden optimal path.

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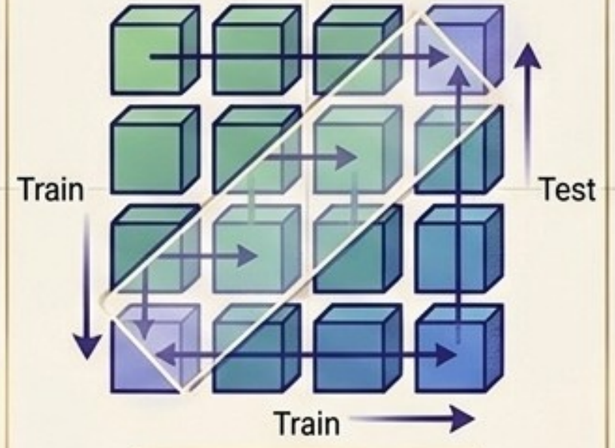
The Machine Learning Universe

The Supervised Curve
(Predictive)



Data-driven modelling

The Validation Matrix
(Robustness)



Validate on unseen data

The RL Feedback Loop
(Adaptive)



Experience-driven adaptation

Master Legend

The Goal:
Uncovering the signal in the noise.

The Methods:
Supervised (Data-driven modelling) or Reinforcement (Experience-driven adaptation).

The Mandate:
Always validate on unseen data and endlessly balance complexity.

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